

BLUETOOTH

***Bluetooth** is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.*

Topics discussed in this section:

Architecture

Bluetooth Layers

Baseband Layer

L2CAP

Figure 14.19 *Piconet*

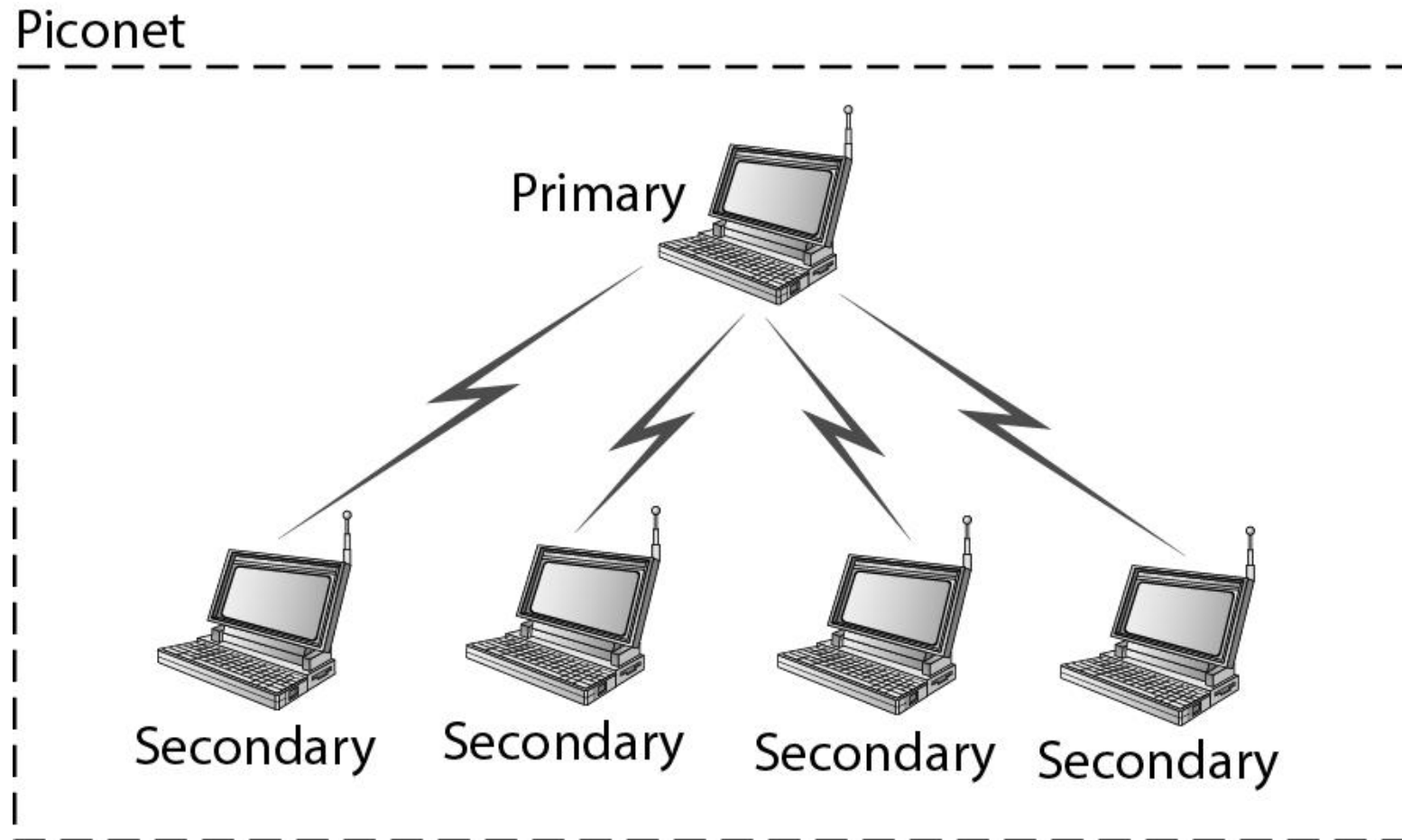


Figure 14.20 *Scatternet*

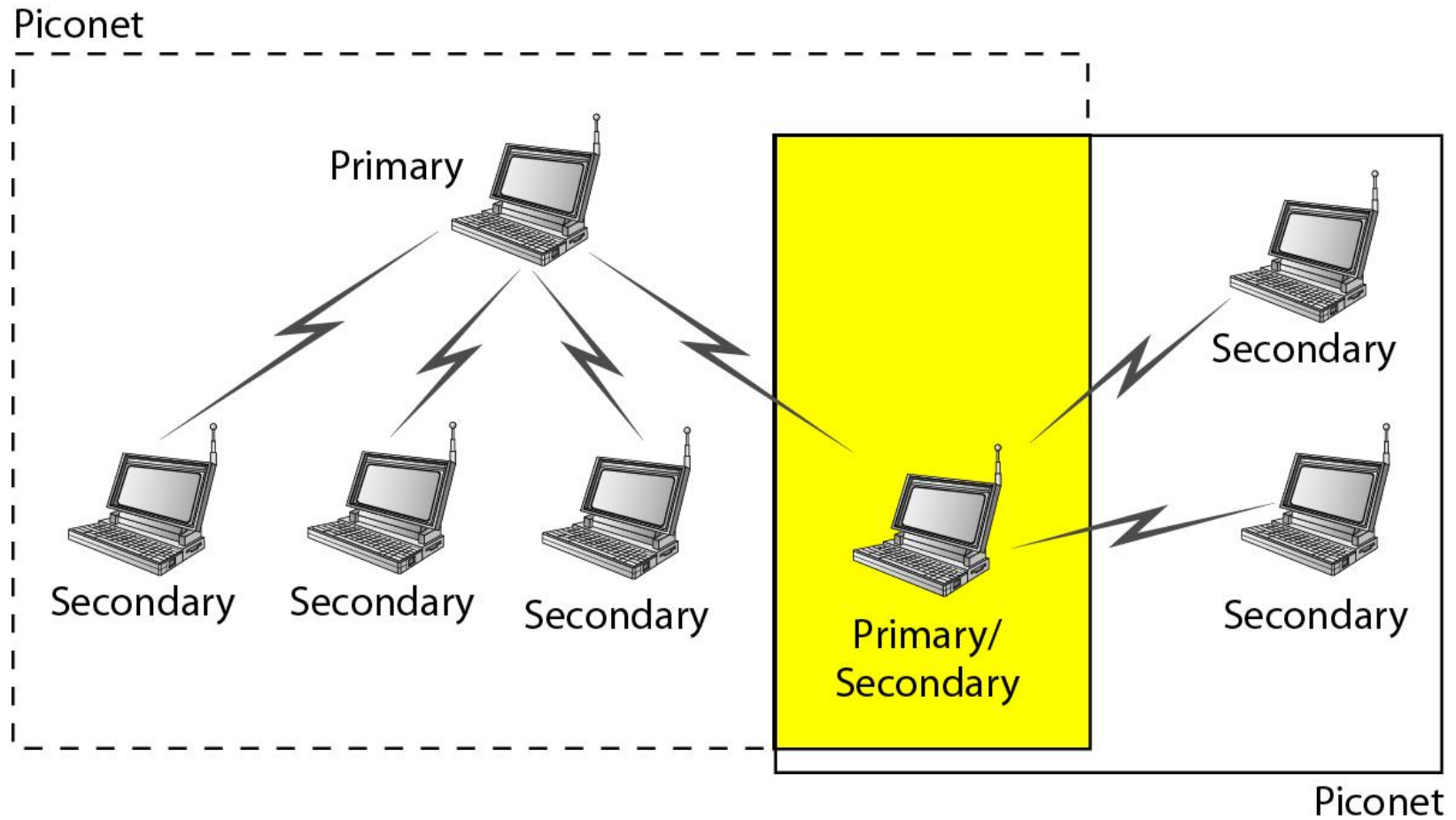
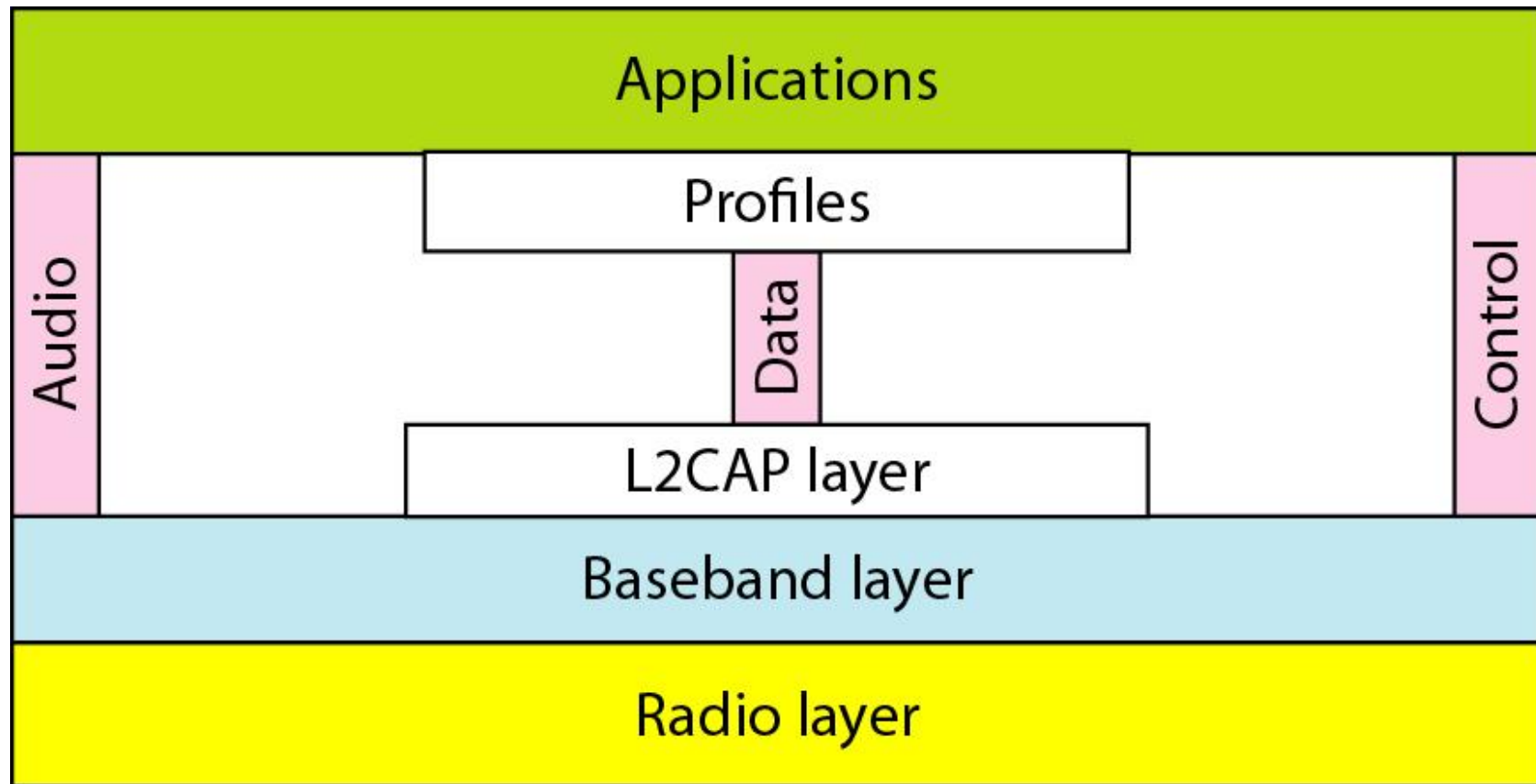


Figure 14.21 *Bluetooth layers*



Conti....

- The Bluetooth Protocol Stack does not follow OSI model and revised by IEEE to fix 802.11.
- The **Physical Radio layer** corresponds to the ISO(International Standardization Organization) physical layer and deals with radio transmission and modulation.
- The **Baseband layer** corresponds to MAC sublayer and some physical layer and handles how the master control time slots.
- The **link manager** handles establishment of logic channel (power management, authentication, QoS).
- The **logical link control adaptation protocol (L2CAP)** shields the upper layer from lower layer
- **Audio and Control protocols** deal with audio and control.
- **LLC (Logical Link Control)** is inserted by IEEE to make it compatible with other 802.
- **RFcomm** (Radio Frequency communication) emulates serial port for connecting mouse, keyboard, modem.
- **Telephony protocol** real-time speech, call setup and termination
- The **service discovery protocol** is used to locate service in a network.
- Each application uses a specific subset of protocols included in **Applications/Profiles** based on the device it is applicable to. Eg mobile or PDA.

Figure 14.25 *L2CAP data packet format*



Figure 14.22 *Single-secondary communication*

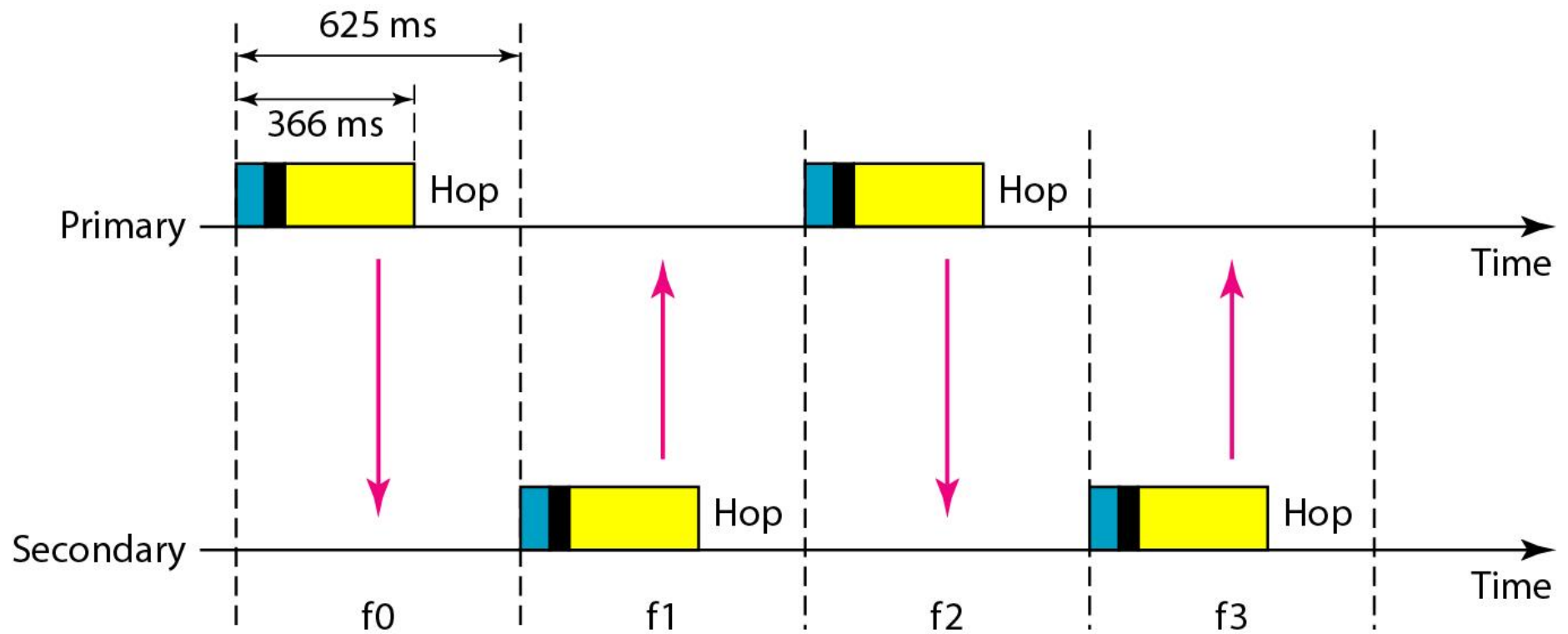


Figure 14.23 *Multiple-secondary communication*

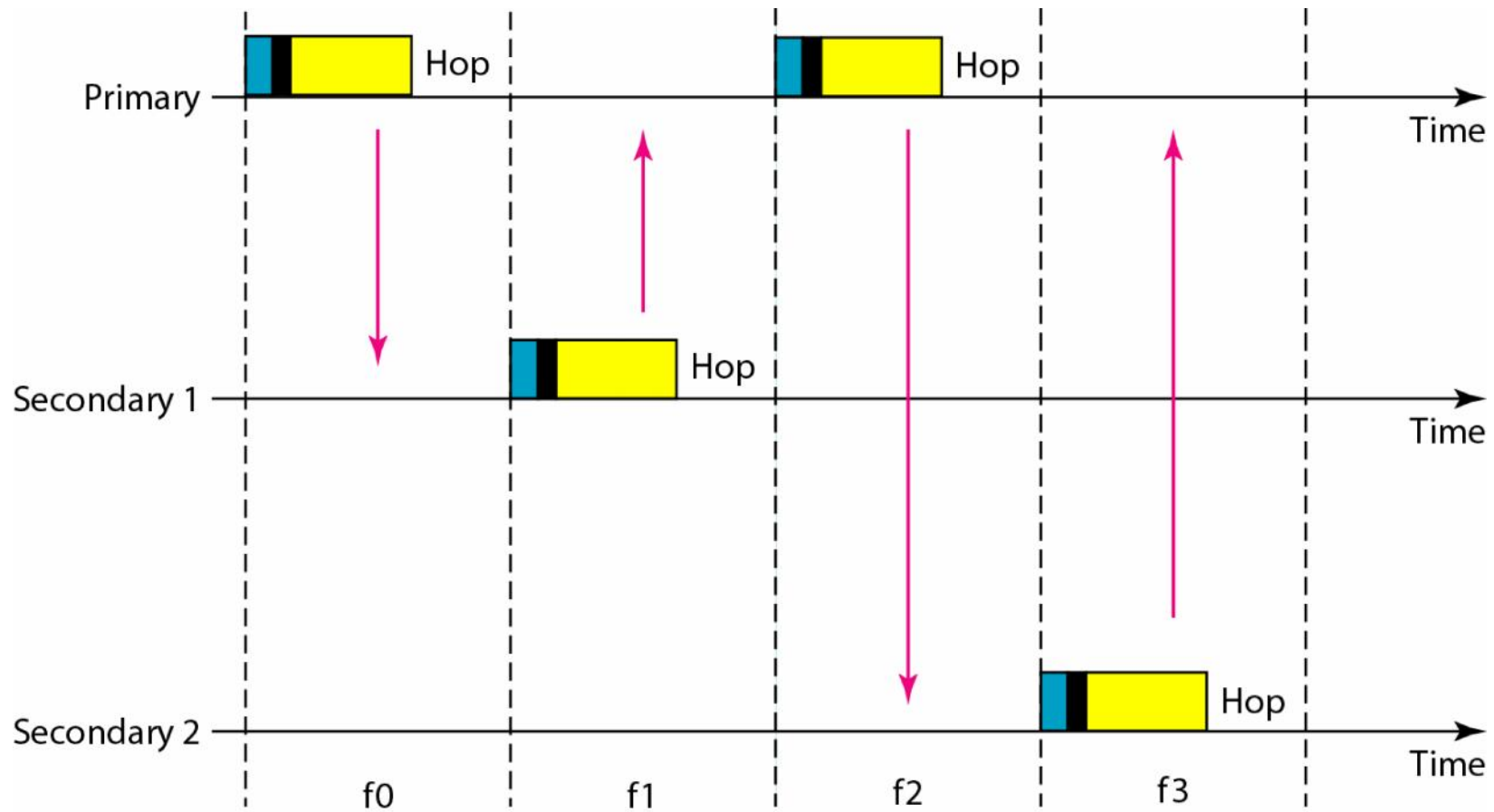
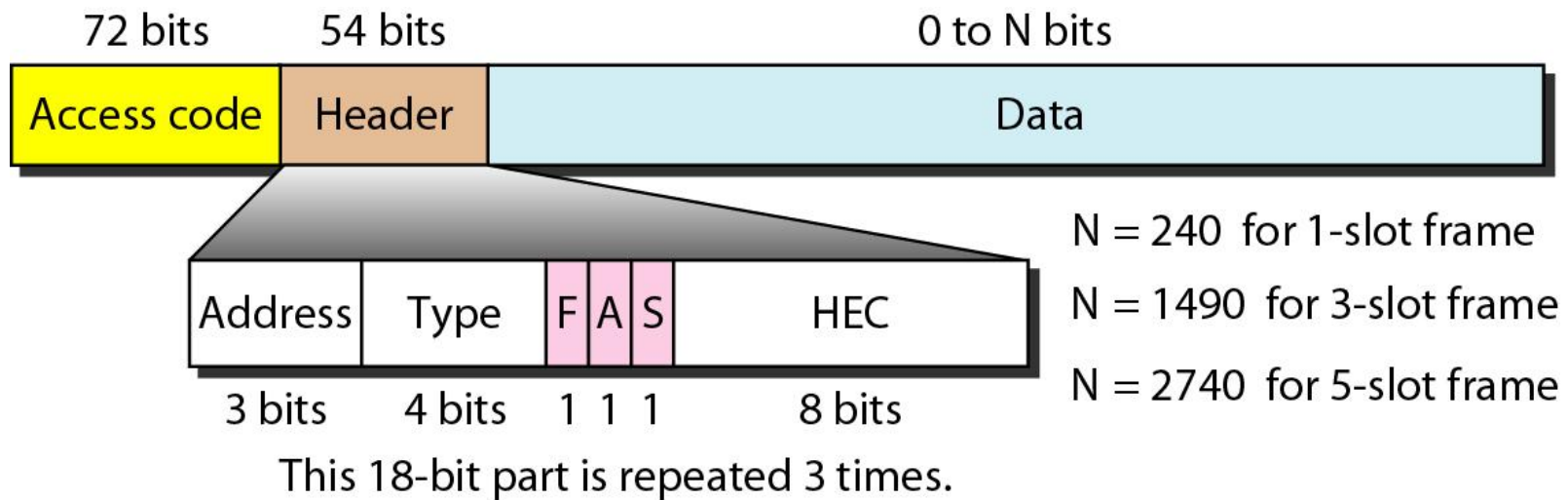


Figure 14.24 *Frame format types*



- Access code. This 72-bit field normally contains synchronization bits and the identifier of the primary to distinguish the frame of one piconet from that of another.
- Header. This 54-bit field is a repeated 18-bit pattern. Each pattern has the following subfields:
 - a. Address. The 3-bit address subfield can define up to seven secondaries (1 to 7).

If the address is zero, it is used for broadcast communication from the primary to all secondaries.
 - b. Type. The 4-bit type subfield defines the type of data coming from the upper layers. We discuss these types later.
 - c. F. This 1-bit subfield is for flow control. When set (1), it indicates that the device is unable to receive more frames (buffer is full).
 - d. A. This 1-bit subfield is for acknowledgment. Bluetooth uses Stop-and-Wait ARQ; 1 bit is sufficient for acknowledgment.
 - e. S. This 1-bit subfield holds a sequence number. Bluetooth uses Stop-and-Wait ARQ; 1 bit is sufficient for sequence numbering.
 - f. HEC. The 8-bit header error correction subfield is a checksum to detect errors in each 18-bit header section.
- Payload. This subfield can be 0 to 2740 bits long. It contains data or control information

Thank You